

Transformers

Attention • Blocks • Encoders & Decoders

How a stack of attention and feed-forward blocks turns token vectors into rich, context-aware representations — explained as a team of doctors reviewing a patient, then mapped onto the 2017 paper.

A TEAM OF DOCTORS REVIEWING A CASE

"The patient has fever,
cough, and pneumonia..."

TOKENIZE + EMBED + POSITION

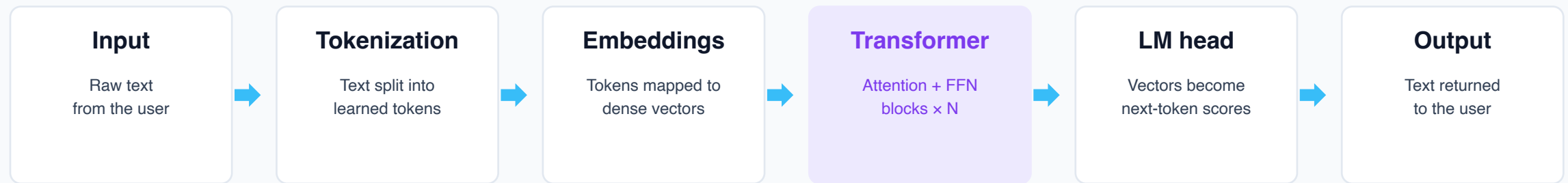
+	-----	+
	ATTENTION (discuss)	
	FFN (think)	
	RESIDUAL + NORM	

+-----+ x N

CONTEXTUAL REPRESENTATION

Where the Transformer Fits in an LLM

Lectures 5–8 covered tokens, embeddings and attention. Today we assemble them into the Transformer itself.



TODAY'S FOCUS

The fourth box. We open it up: what happens inside one Transformer block, why we stack many of them, and how the original 2017 encoder–decoder design became today's decoder-only LLMs.

THE RUNNING ANALOGY

A hospital receives a case. A team of doctors — each with a different role — reviews and refines it again and again. Every Transformer component maps to something this team does.

Analogy: One Patient, One Team of Doctors

“The patient has fever, cough, and a history of pneumonia. The patient was started on antibiotics.”

A team of doctors

Each one has a different role — like different parts of the network.

Repeated review

They read the case again and again — like stacked Transformer blocks.

A refined picture

Each round sharpens their understanding — a richer representation.

KEY STATEMENT

Think of a Transformer as a team of specialists repeatedly reviewing a document. Every slide today maps one component to something this team does.

Tokenization: Break the Case into Pieces

IN THE HOSPITAL

- A long medical report is too big to handle as one object.
- First step: split it into individual facts —
- patient · age · symptoms · diagnosis · medication · labs

IN THE TRANSFORMER

- Text is split into tokens — the units the model reads.
- Real tokenizers often split words into sub-word pieces.
- Text → Tokenization → Tokens (Lecture 5)

EXAMPLE

```
"The patient has fever, cough, history of pneumonia"  
-> ["The", "patient", "has", "fever", "cough", "history", "pneumonia", ...]
```

For intuition, treat each token as one fact from the report. The exact split depends on the tokenizer.

Embedding: Give Each Piece a Profile Card

IN THE HOSPITAL — A PROFILE CARD

fever

- symptom
- temperature-related
- medical · commonly linked to infection

IN THE TRANSFORMER — A VECTOR

```
fever → [0.21, -0.73, 0.41, ...]
```

The computer can't store “symptom” or “medical”. Each token becomes a learned numerical profile.

IMPORTANT — THE PROFILE IGNORES THE SENTENCE

“bank” starts with the same embedding in “money was deposited in the bank” and “we sat beside the bank of the river”. Fixing that is attention's job.

patient → patient profile



fever → symptom profile



cough → respiratory profile



drug → medication profile

Positional Information: A Seat and a Time

IN THE HOSPITAL — THE PATIENT TIMELINE

Day 1

fever

Day 2

cough

Day 3

antibiotics

Day 5

improvement

Shuffle the days and the story changes.

IN THE TRANSFORMER — WORD ORDER

Dog bites man
≠ Man bites dog

Same words, different meaning. Position tells the model where each token occurs.

token + position = position-aware vector

Seat 1
The



Seat 2
patient



Seat 3
has



Seat 4
fever



Seat 5
cough

Without position information, attention treats the input as an unordered bag of tokens.

Attention: Who Should “bank” Listen To?

“I deposited money in the bank.”

FOCUS ON

money · deposited

MEANING

a financial institution

“We walked along the bank of the river.”

FOCUS ON

walked · river

MEANING

the side of a river

IN THE HOSPITAL

A doctor asks: “Which other pieces of information in this case are relevant to understanding this one?”

KEY STATEMENT

Attention determines which other tokens should influence the current token's representation.

Query, Key, Value

Q · QUERY

“What am I looking for?”

bank asks: what do I need to understand my role in this sentence?

$$Q = X \cdot W_Q$$

K · KEY

“What kind of information do I have?”

money → financial
river → geographical
drug → medication

$$K = X \cdot W_K$$

V · VALUE

“What can I actually give you?”

river's Key: “I'm geographical.”
Its Value: the location context itself.

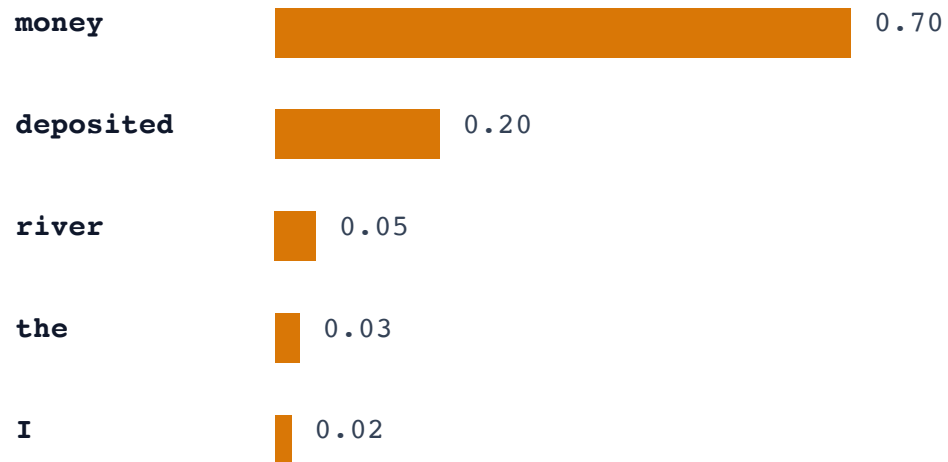
$$V = X \cdot W_V$$

THE KEY DISTINCTION

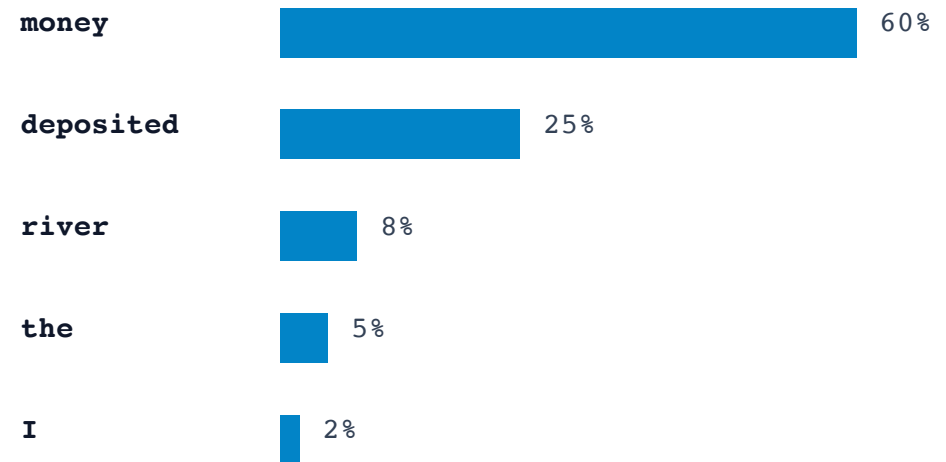
The Key decides WHETHER to attend. The Value is WHAT gets passed along. Compatibility of a query with every key: $Q \cdot K^T$.

Scores and Softmax: Dividing 100% of Attention

RELEVANCE SCORES ($Q \cdot K^T$)



AFTER SOFTMAX (sums to 100%)



SOFTMAX

Converts raw scores into non-negative weights that sum to 1 — like a doctor splitting 100% of their attention across colleagues. (Numbers are illustrative.)

Weighted Values: Combine Every Expert's Input

THE DOCTOR DOESN'T PICK JUST ONE

60% × value(money)

25% × value(deposited)

8% × value(river)

5% × value(the)

2% × value(I)

= a new, context-aware “bank”

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \cdot V$$

$Q \cdot K^T$

compatibility of each query with each key

$/\sqrt{d_k}$

keeps scores in a stable range

softmax

scores become weights that sum to 1

$\cdot V$

weighted sum of the values

The output for each token is a blend of every token's Value, weighted by attention: a new, context-aware representation.

Self-Attention: One Team Talking Among Itself

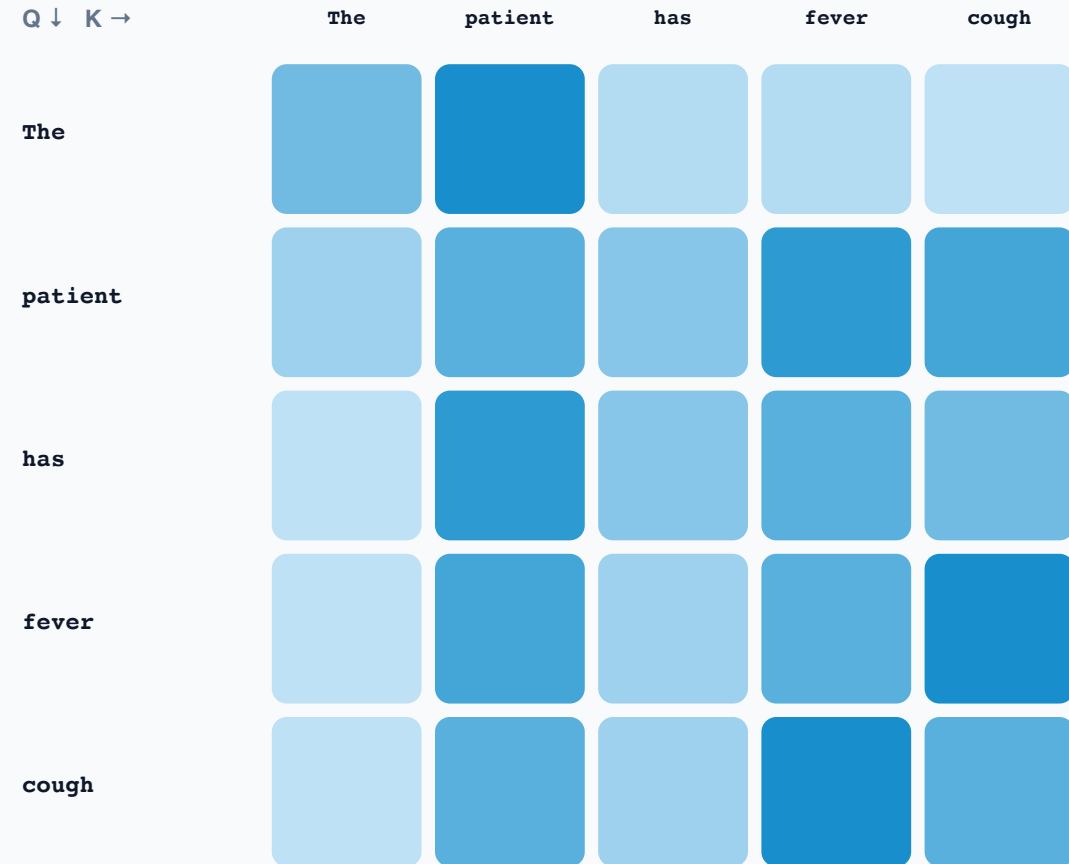
Q, K and V all come from the same sequence.

Doctors on the same team consult each other: every token can attend to every other token, including itself.

READING THE GRID

Each row is one token's query; each column a key it can look at.
Darker = more attention (illustrative).

Q ↓ K →



Multi-Head Attention: A Panel of Specialists

IN THE HOSPITAL — SAME PATIENT, DIFFERENT FOCUS

Cardiologist

Neurologist

Radiologist

Oncologist

Pharmacologist

IN THE TRANSFORMER — CONCEPTUALLY

Head 1 → syntactic relationships

Head 2 → entity relationships

Head 3 → long-distance relationships

Head 4 → semantic relationships

```
head_i = Attention(Q·W_Qi, K·W_Ki, V·W_Vi)
MHA    = Concat(head_1, ..., head_h) · W_O
```

CAVEAT

Heads aren't guaranteed to match human specialties — each learns its own projections and attention patterns during training.

Multiple Perspectives: Four Heads, One Sentence

“The patient was given aspirin because he had chest pain.”

HEAD 1

patient ↔ given

who received it

HEAD 2

aspirin ↔ given

what was given

HEAD 3

chest pain ↔ because

the reason

HEAD 4

he ↔ patient

who “he” refers to

An intuition, not a literal description of what every head learns. Ask: which head would you need to resolve “he”?

Feed-Forward Network: Talk Together, Then Think Alone

ATTENTION

“Talk to everyone relevant.”

Doctors exchange information. Tokens mix with each other.



FFN / MLP

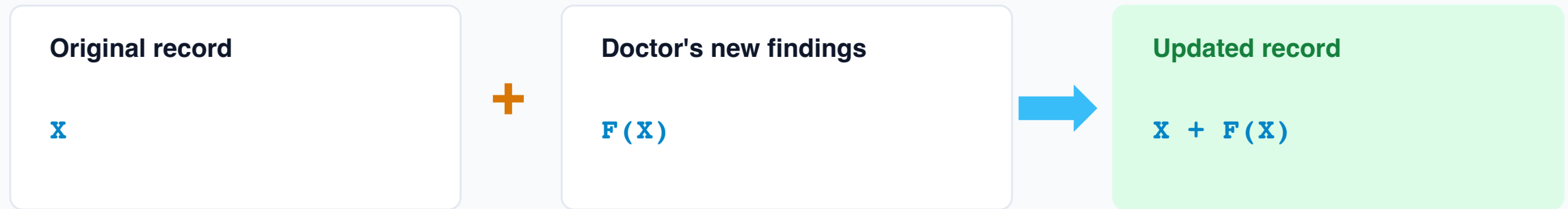
“Now process what I learned.”

Each doctor works on their own notes. Each token is transformed independently.

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \cdot \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$

Attention is the only place tokens communicate. The FFN is a nonlinear transformation applied to each position separately.

Residual Connection: Never Throw Away the Record



$$\text{Output} = X + F(X)$$

The sublayer only has to learn what to ADD. The original information keeps flowing through — which also helps deep stacks train.

IN THE HOSPITAL

After a review you don't discard the original record — you keep it and attach the new findings.

Layer Normalization: Standardize the Measurements

IN THE HOSPITAL — FOUR DOCTORS, FOUR SCALES

```
Doctor A  ->  0.001
Doctor B  ->  5000
Doctor C  -> -300
Doctor D  ->  0.02
```

Put them on one scale before combining.

IN THE TRANSFORMER

Keeps internal representations numerically stable and easier to optimize.

LayerNorm — original 2017 Transformer

RMSNorm — common in modern LLMs

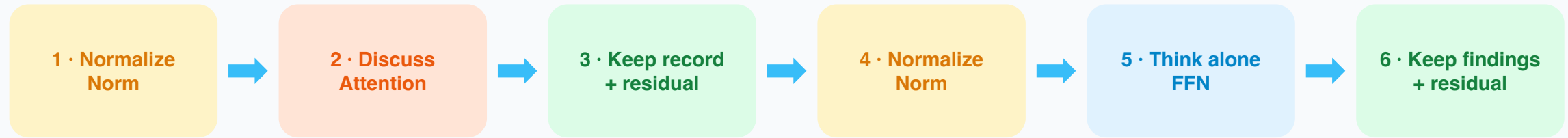
Avoid the Confusion: Position ≠ Normalization

Component	Analogy	Purpose
Positional encoding / RoPE	Patient timeline	Where is this token?
Normalization (LayerNorm / RMSNorm)	Standardizing measurements	Keep representations numerically well behaved

KEY STATEMENT

They solve completely different problems: one answers “when?”, the other keeps the arithmetic stable.

A Transformer Block = One Review Round



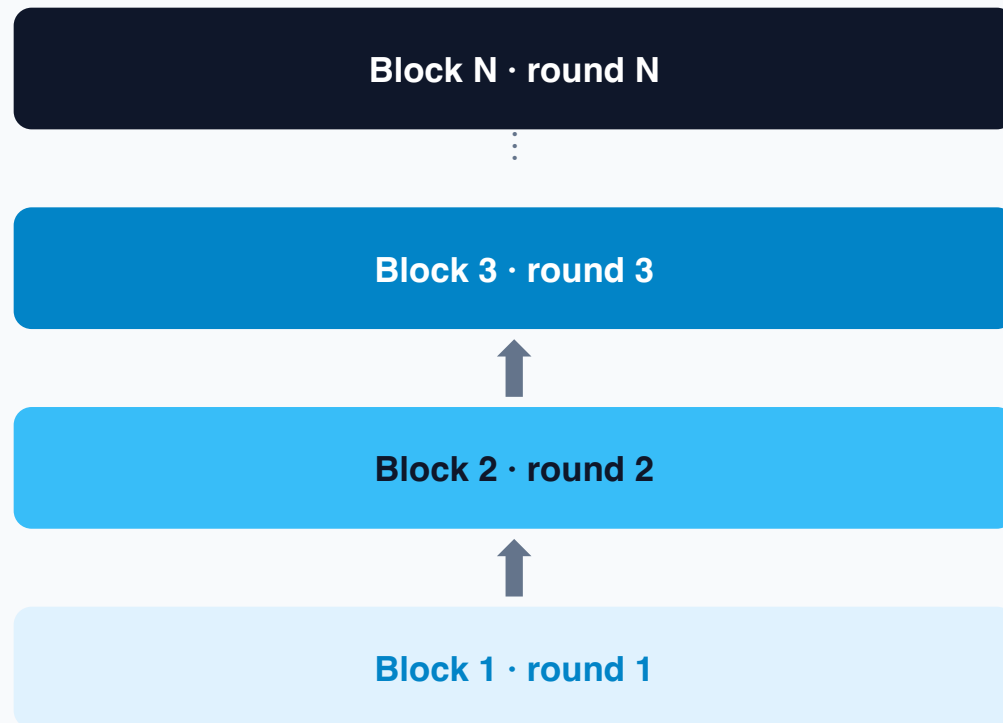
```
Y = X + Attention(Norm(X))  
Z = Y + FFN(Norm(Y))
```

A modern pre-norm block: normalize before each sublayer, then add the result back to the running record.

IN THE HOSPITAL

Normalize the information → doctors discuss → keep the original + findings → each doctor thinks → keep original + findings → updated patient representation.

Stacking Blocks: Many Rounds of Review



One review isn't enough for a complicated case.

Each block takes the previous representation and refines it further — same structure, different learned weights.

IN REAL MODELS

32, 40, 80 or more blocks are stacked.

Causal Masking: No Reading Future Pages

When writing page 2 of a report, the doctor can't secretly read page 4.

Predicting “severe” in “The patient developed severe fever”, the model sees The · patient · developed · severe — but not “fever”.

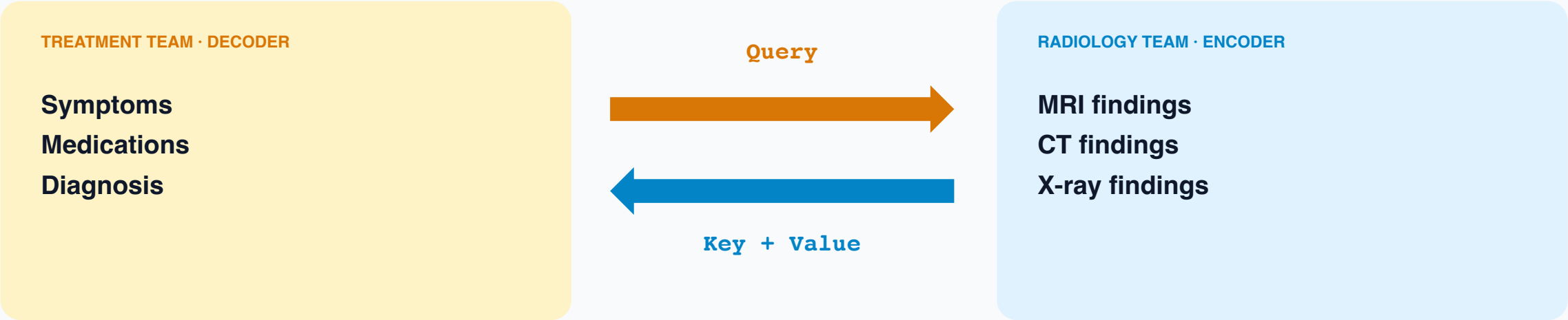
Look backward, never forward.

Blue = can attend · grey = masked (future)

Q ↓ K →

	The	patient	develop.	severe	fever
The					
patient					
develop.					
severe					
fever					

Cross-Attention: One Team Consults Another



Attention type	Where Q, K, V come from
Self-attention	Q, K and V all from the same sequence
Cross-attention	Q from the decoder; K and V from the encoder

The Encoder Reads; the Decoder Writes

ENCODER · MEDICAL REVIEW TEAM

Reads the whole case before producing its representation.

```
The <-> patient <-> fever <-> cough
```

Every token sees every other token → contextual representations.

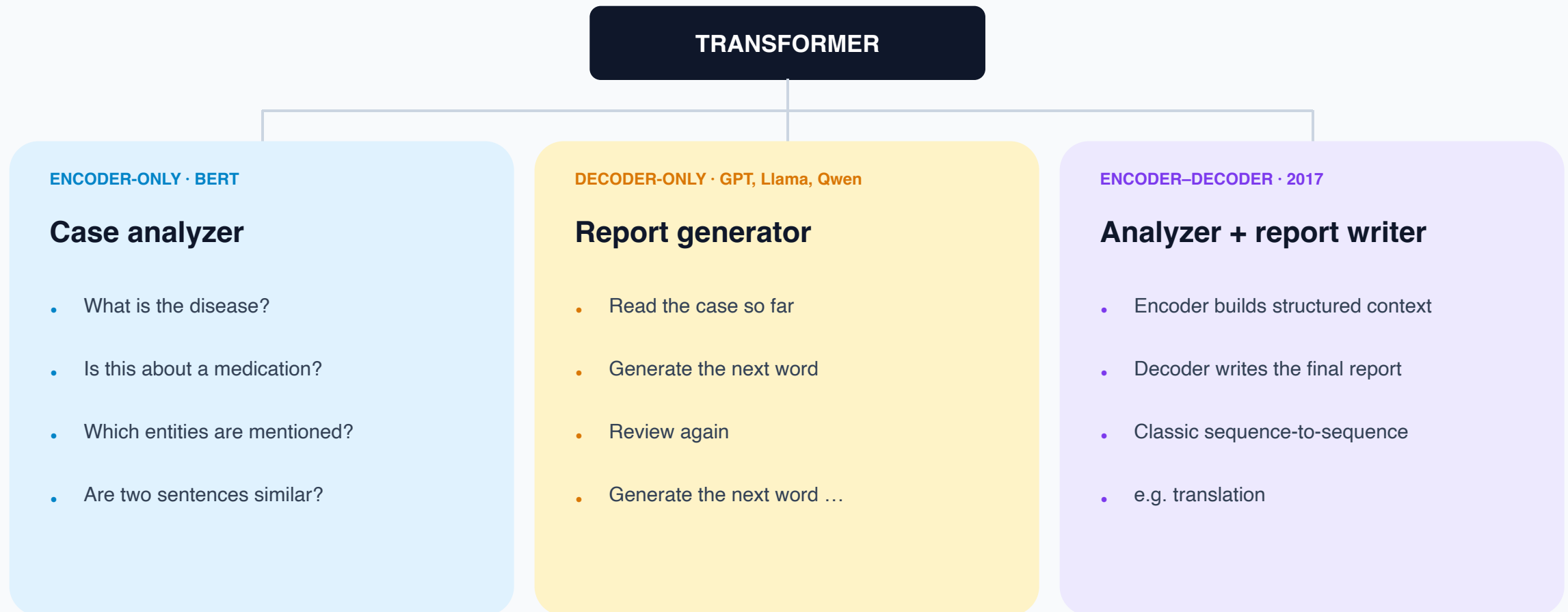
DECODER · REPORT WRITER

Writes one word at a time.

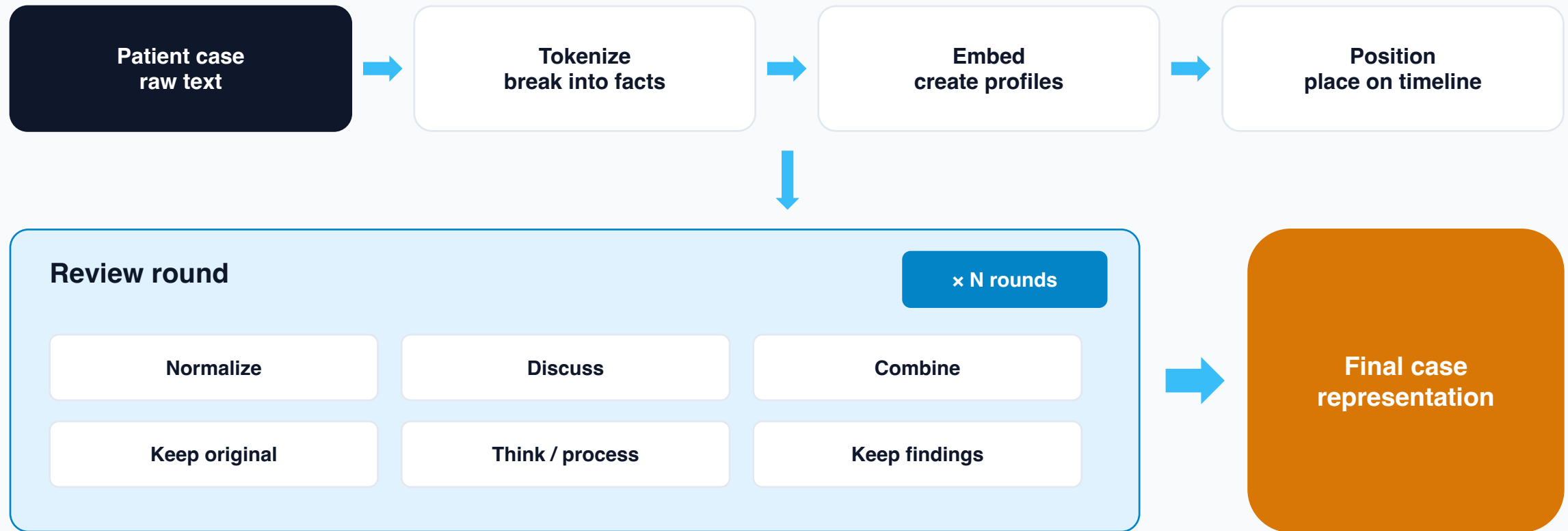
```
Diagnosis:  
Diagnosis: pneumonia  
Diagnosis: pneumonia with  
Diagnosis: pneumonia with bacterial ...
```

Each new word depends on what's already written — autoregressive.

Three Ways to Assemble the Team



The Whole Hospital, End to End



This is the mental model to keep: tokenize, embed, add position, then N rounds of review.

Component → Analogy → Job

Component	Real-life analogy	What it does
Tokenization	Break a medical report into individual facts	Splits text into tokens
Embedding	Create a profile card for each fact	Converts tokens to vectors
Position	Patient timeline / event order	Represents sequence position
Query	“What information do I need?”	What a token is looking for
Key	“What kind of information do I have?”	Determines relevance
Value	“What information can I provide?”	Information passed along
Attention	Doctors discussing relevant facts	Selectively gathers information
Self-attention	Doctors in the same team consulting each other	Q, K, V from the same sequence
Multi-head attention	Multiple medical specialists	Multiple learned projections
Softmax	Divide 100% attention among experts	Converts scores into weights

Component → Analogy → Job

Component	Real-life analogy	What it does
Weighted sum	Combine expert opinions	Contextual representation
FFN / MLP	Doctor analyzing gathered evidence alone	Nonlinear transformation
Residual connection	Keep original record + new findings	Preserves information
Normalization	Standardize measurements / reports	Stabilizes activations
Causal mask	Cannot read future pages	Prevents looking ahead
Cross-attention	Treatment team consulting radiology	Connects two sequences
Transformer block	One complete case-review round	Attention + FFN + residual + norm
Multiple blocks	Multiple rounds of clinical review	Deep representation learning
Encoder	Case-analysis team	Builds input representation
Decoder	Report-writing team	Generates output sequentially

If Students Remember One Thing

A Transformer is a team of specialists repeatedly reviewing a document.

Embedding

Give every word a profile.

Position

Give every word a place in the document.

Attention

Let each word look at other relevant words.

Q / K / V

Looking for? · Have? · Can give?

Multi-head

Several specialists, different perspectives.

FFN

Each specialist processes what they gathered.

Residual

Never throw away the record; add findings.

Normalization

Keep measurements stable and standardized.

Layers

Repeat the review many times.

Decoder

Write the response one token at a time.

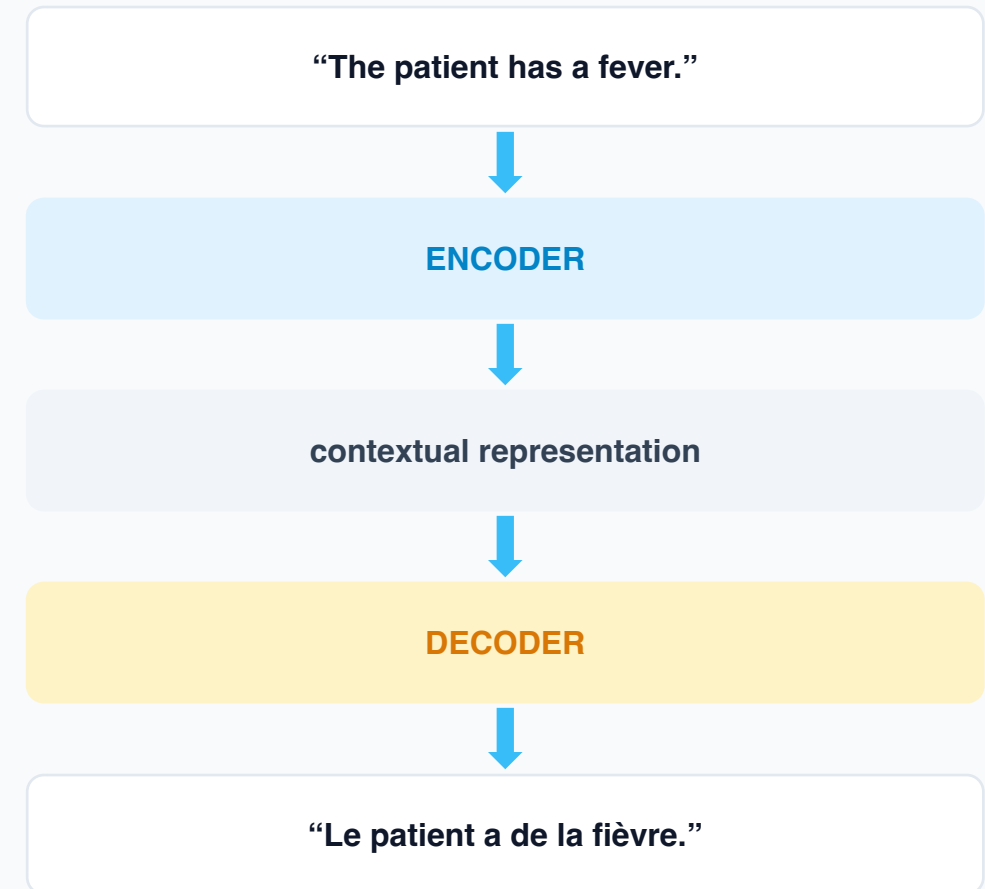
The Original Transformer — Attention Is All You Need

Vaswani et al., 2017

Introduced for sequence-to-sequence tasks — especially machine translation.

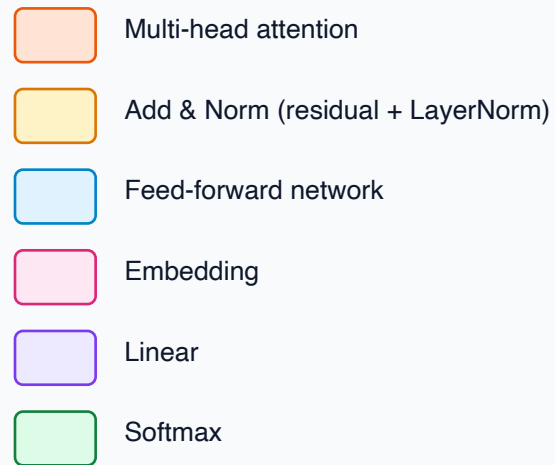
KEY POINT

It was NOT designed as a decoder-only LLM. It was Encoder + Decoder
→ sequence-to-sequence prediction.



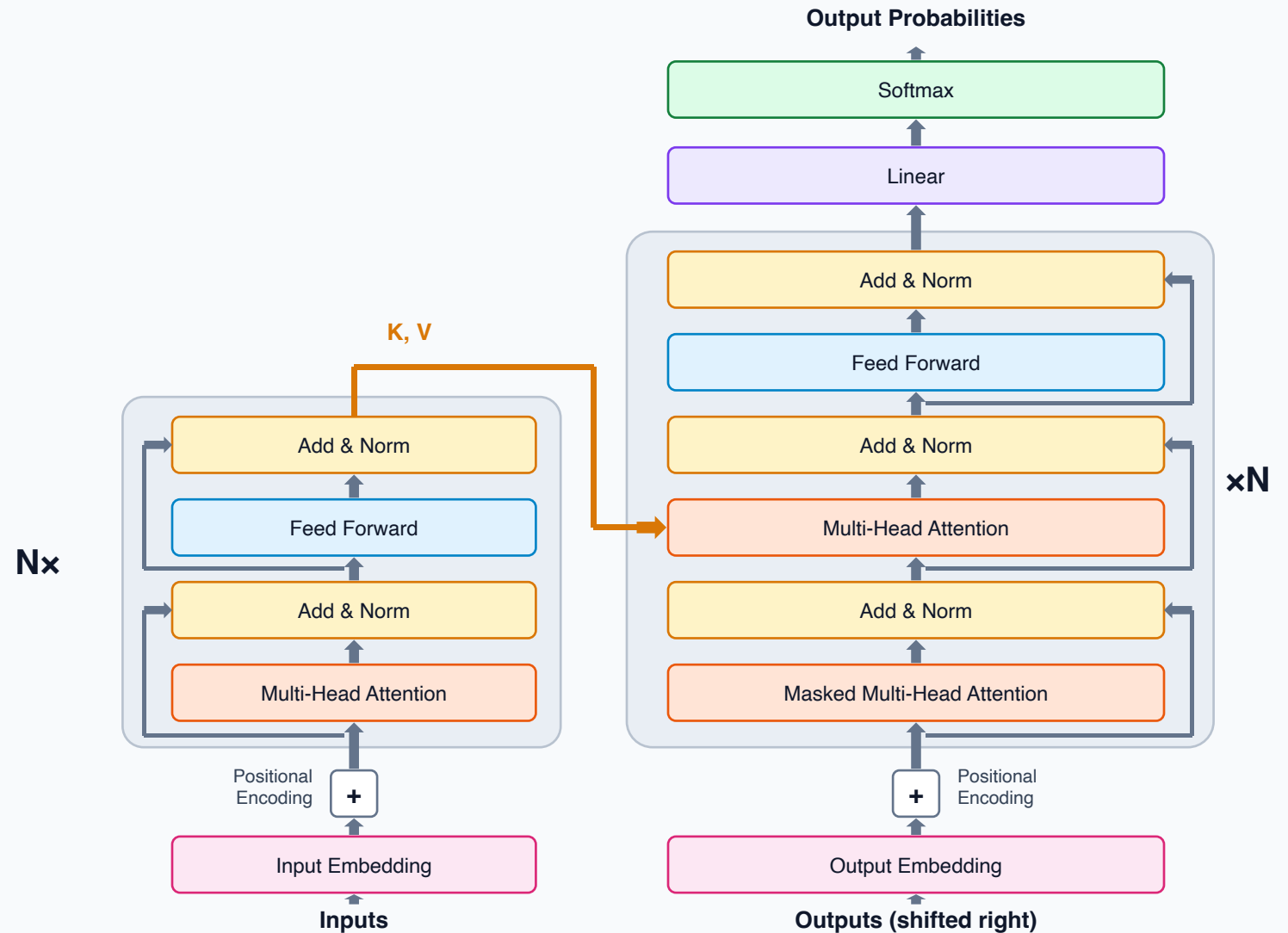
The Original Transformer Architecture

Encoder on the left, decoder on the right. Data flows upward.
Each grey frame is one layer, repeated $N = 6$ times.

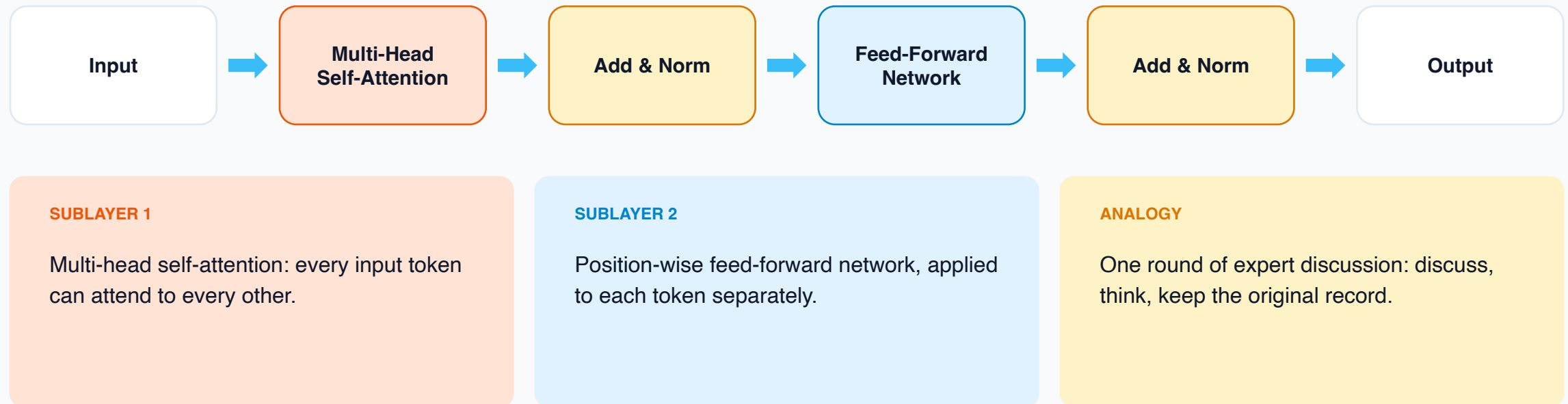


CROSS-ATTENTION

The orange line carries the encoder's output into the decoder: K and V from the encoder, Q from the decoder.

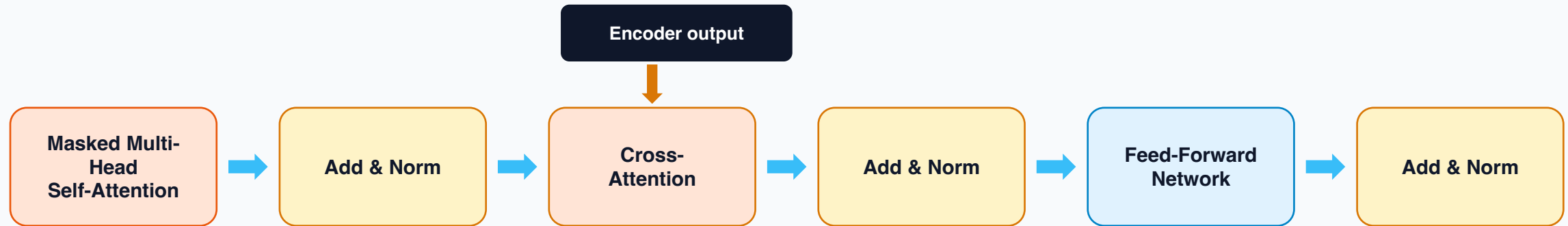


Original Encoder Layer: Two Sublayers (N = 6)



Each sublayer is wrapped in a residual connection followed by layer normalization — “Add & Norm”. Six such layers are stacked.

Original Decoder Layer: Three Sublayers (N = 6)



Why three? The decoder has two jobs.

1 · LOOK AT WHAT IT HAS WRITTEN

→ masked self-attention: earlier output tokens only.

2 · LOOK AT THE INPUT SEQUENCE

→ cross-attention over the encoder's output.

Where Q, K and V Come From in the Original Paper

Attention	Q from	K, V from	Can see
Encoder self-attention	Encoder	Encoder	Every input token
Decoder masked self-attention	Decoder	Decoder	Earlier output tokens only
Decoder cross-attention	Decoder	Encoder	Every input token

Decoder

--Q-->

Cross-Attn

Encoder

--K,V->

Cross-Attn

Cross-attention: the decoder asks questions about the encoder's representation.

Add & Norm in the Paper's Diagram

ADD = RESIDUAL CONNECTION

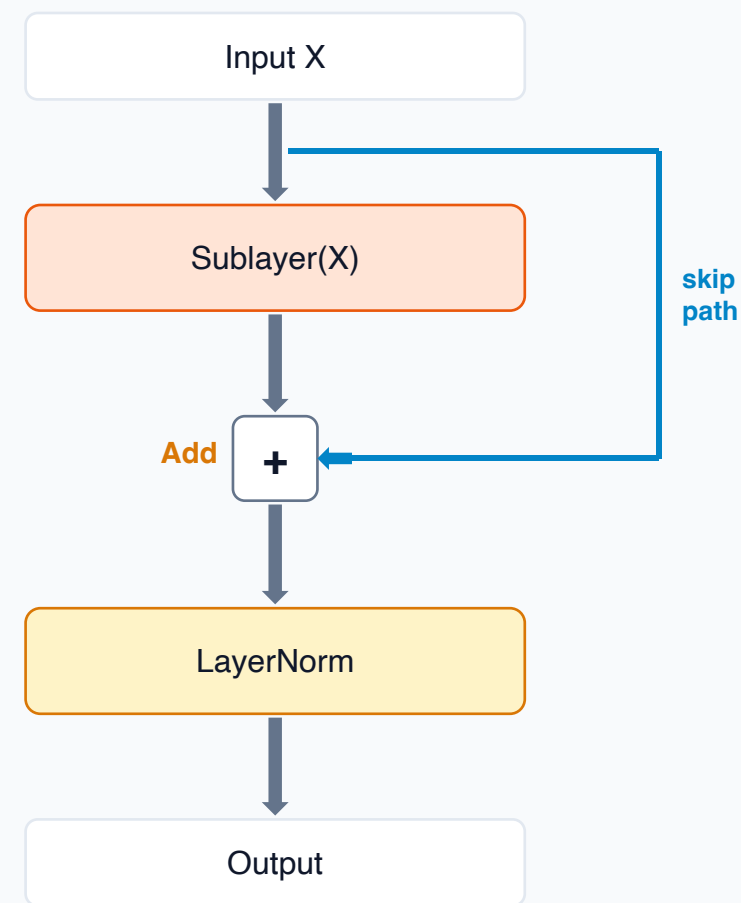
```
Output = X + Sublayer(X)
```

NORM = LAYER NORMALIZATION

```
LayerNorm( X + Sublayer(X) )
```

ORIGINAL VS MODERN

The paper normalizes AFTER the residual (post-norm) and uses LayerNorm. Many modern LLMs normalize BEFORE each sublayer (pre-norm) and use RMSNorm.



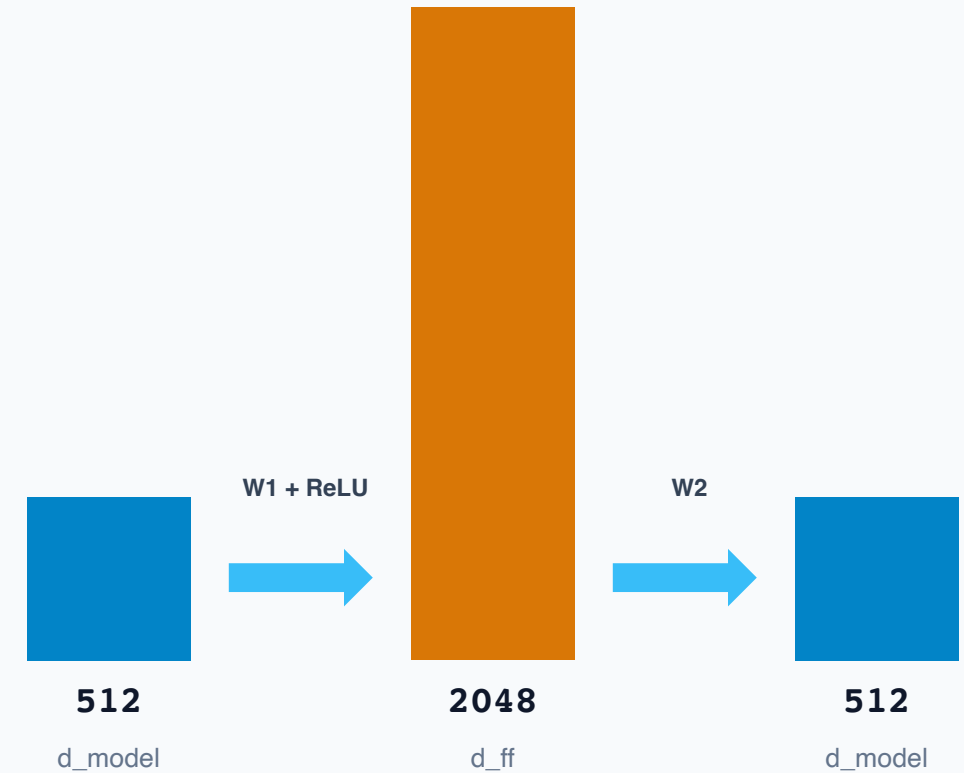
Position-wise Feed-Forward Network in the Paper

$$\text{FFN}(\mathbf{x}) = \max(0, \mathbf{x} \cdot \mathbf{W1} + \mathbf{b1}) \cdot \mathbf{W2} + \mathbf{b2}$$

Applied independently and identically to each position. Activation: ReLU —
Linear → ReLU → Linear.

MODERN LLMs

Often much larger MLPs, with gated activations such as SwiGLU.

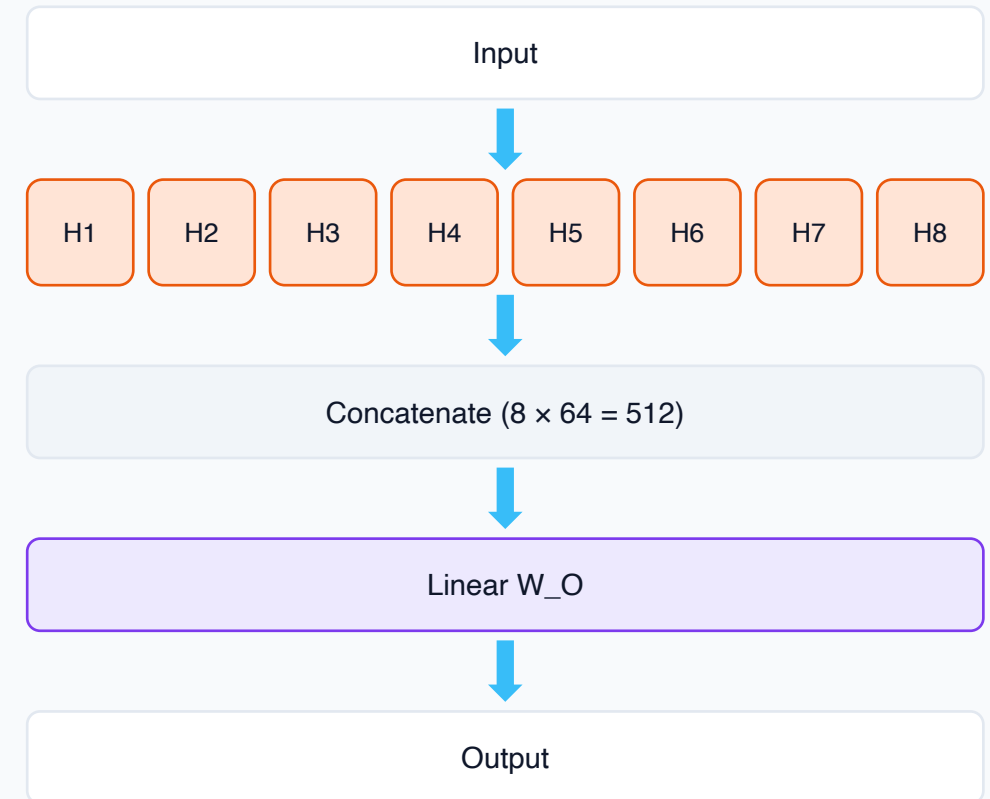


Multi-Head Attention in the Paper: $h = 8$

```
MultiHead(Q,K,V) = Concat(head_1..head_8)·W_O  
head_i = Attention(Q·W_Qi, K·W_Ki, V·W_Vi)
```

512 d_{model} **8**heads (h)**64** $d_k = d_v$ **$8 \times 64 = 512$**

Not eight separate Transformers: one representation projected into 8 learned subspaces, with attention in each.



Positional Encoding in the Paper: Sinusoidal

```
PE(pos, 2i)    = sin( pos / 10000^(2i / d_model) )  
PE(pos, 2i+1) = cos( pos / 10000^(2i / d_model) )
```

Input = Embedding + PositionalEncoding

EMBEDDING

Who is speaking?

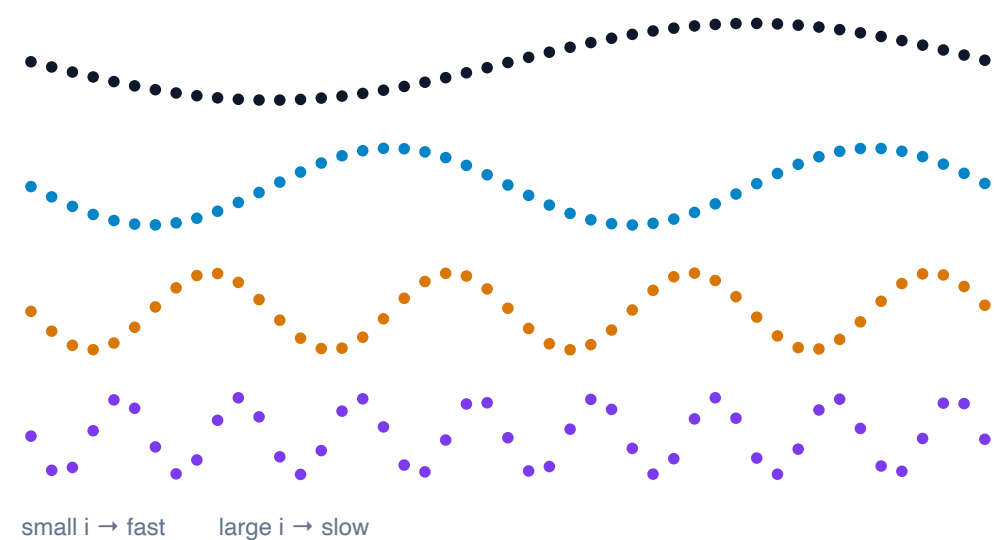
POSITION

When / where did they speak?

NO RoPE YET

The original Transformer used sinusoidal encodings. Many modern LLMs use RoPE instead.

EACH DIMENSION PAIR HAS ITS OWN FREQUENCY

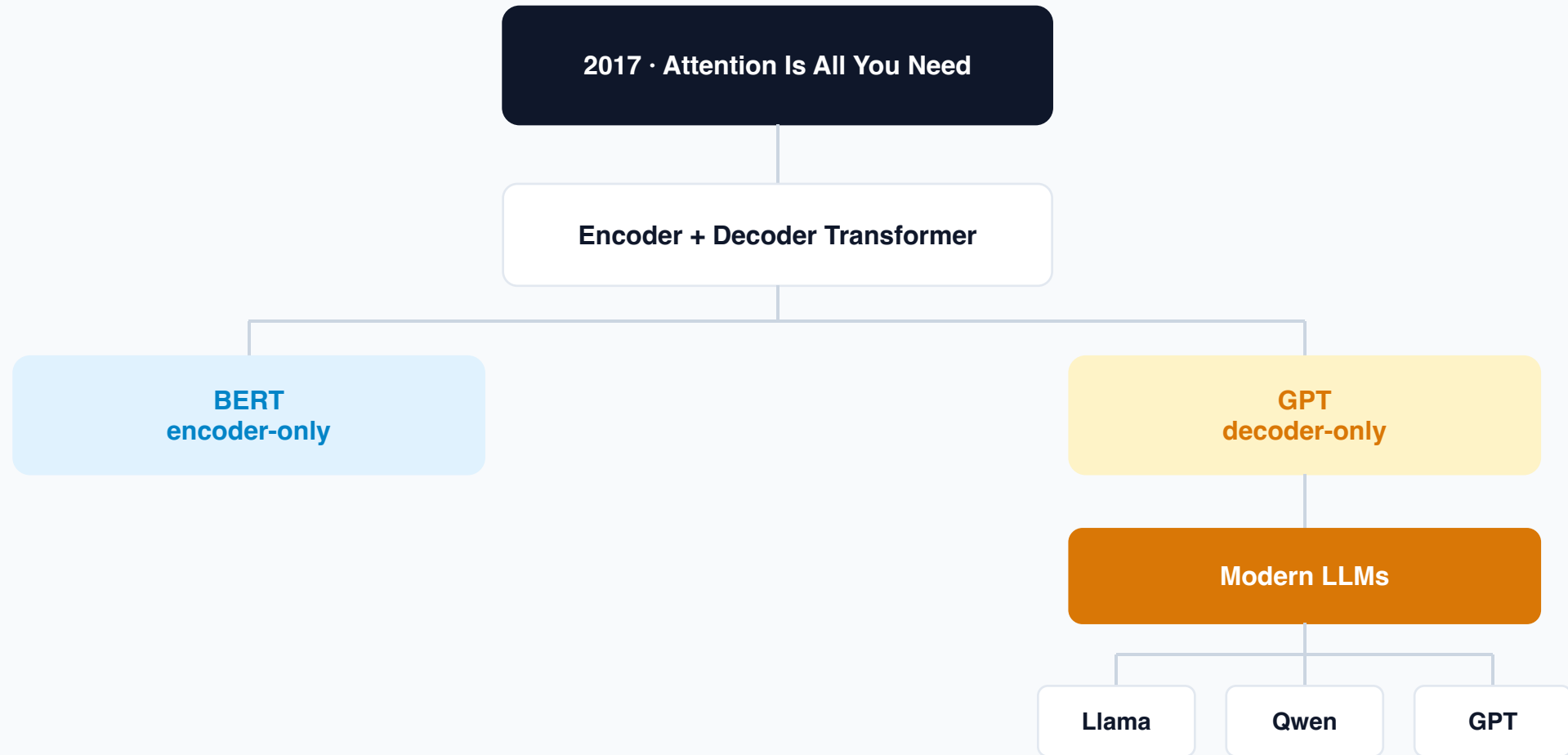


Original Transformer vs Modern Decoder-only LLM

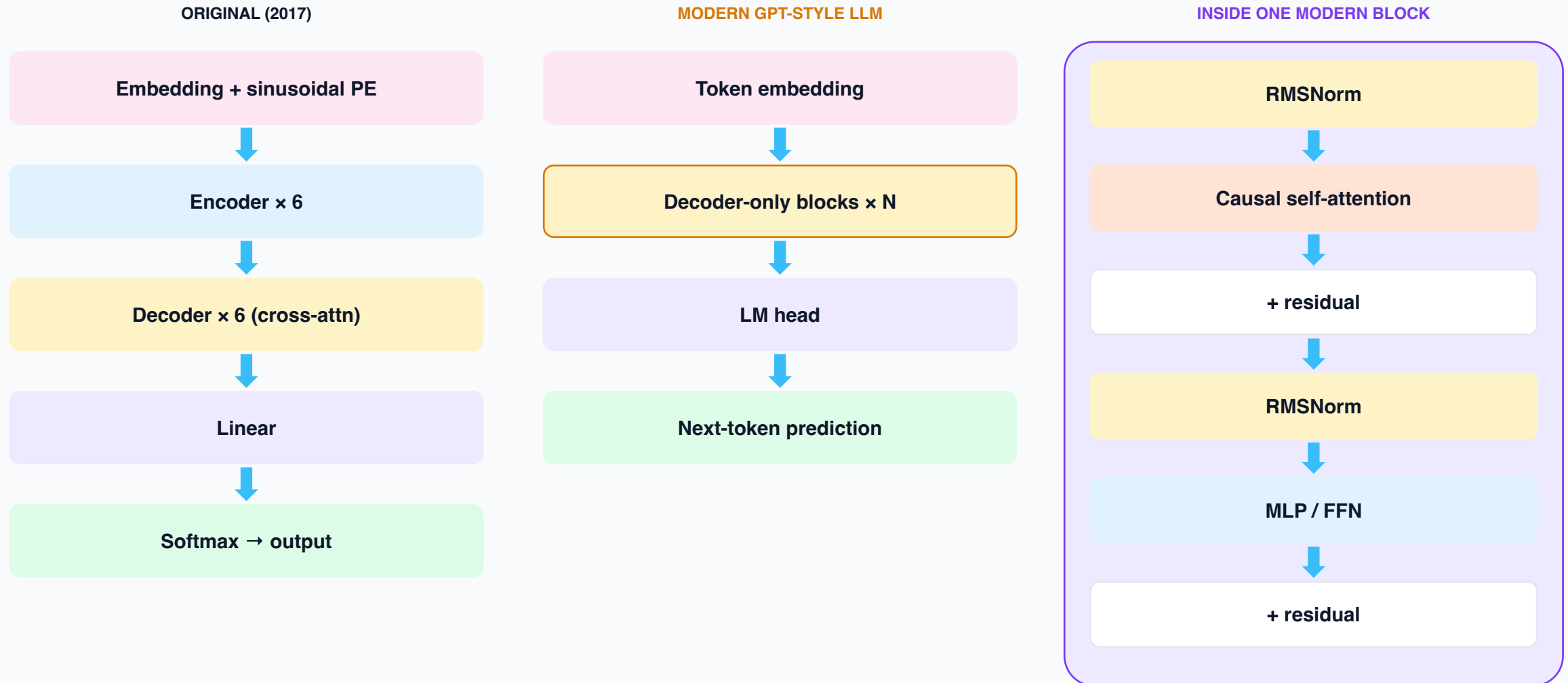
Aspect	Original (2017)	Modern decoder-only LLM
Structure	Encoder + decoder	Decoder only
Original job	Machine translation	General text generation
Layers	6 encoder + 6 decoder	Often dozens of decoder layers
Attention heads	8	Architecture-dependent
d_model	512	Often thousands
FFN activation	ReLU	Often SwiGLU / SiLU-family
Position	Sinusoidal encoding	Often RoPE or other schemes
Normalization	LayerNorm	Often RMSNorm
Cross-attention	Yes	Generally none (no encoder)
Generation	Autoregressive decoder	Autoregressive

GPT-style LLMs didn't invent the Transformer — they adapted its decoder stack and scaled it enormously.

The Evolution: From One Paper to Today's LLMs

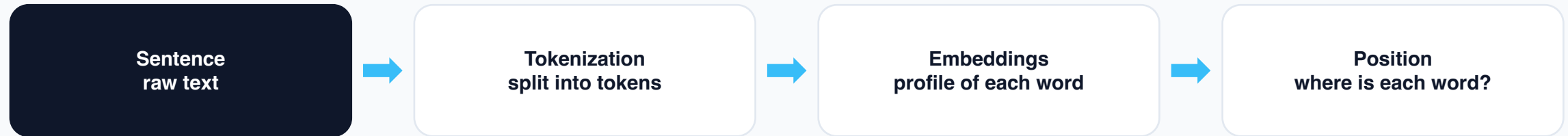


The 2017 Diagram Is Not Today's LLM

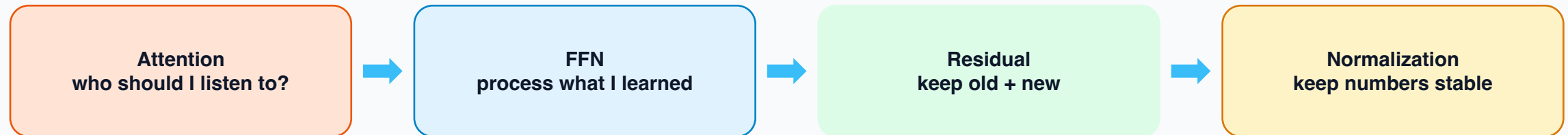


The Big Picture: “The bank approved the loan”

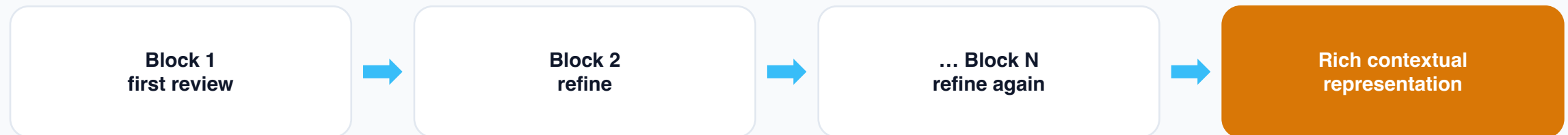
1 · INPUT



2 · INSIDE EACH BLOCK



3 · STACKED



Next Lecture: Language Modeling

The Transformer has built a rich representation of the context. What do we do with it?



BRIDGE

That is where language modeling and next-token prediction begin.